Comprehensive Analysis of Skill Requirements for Data Science Positions in Vietnam: A Novel Framework using Multilingual Topic Modeling and Implications for AI Adoption in Higher Education

Ha The Anh, Nguyen Hoang Anh, Bach Quang Tung
Faculty of Information Technology
HUTECH University
Ho Chi Minh City, Vietnam
Emails: hatheanh, hoanganh, quangtung@student.hutech.edu.vn

Le Nhat Tung
Faculty of Information Technology
HUTECH University
Ho Chi Minh City, Vietnam
lenhattung@hutech.edu.vn

Abstract

The digital transformation era, driven by the explosive growth of Big Data and Artificial Intelligence (AI), has fundamentally reshaped the global labor market. In developing economies like Vietnam, this trend is particularly pronounced, creating a dynamic yet challenging environment for recruitment. A critical issue emerging from this rapid evolution is the persistent "skills gap" between the static nature of academic curricula and the fluid, real-time requirements of the industry. Traditional methods of labor market analysis, such as surveys and manual expert consultations, are increasingly inadequate due to their high cost, latency, and inability to capture the nuances of massive, unstructured data.

To address these challenges, this paper proposes a robust, automated, and data-driven framework to mine recruitment trends directly from online job postings (JDs). We compiled a comprehensive, domain-specific dataset of 3,125 job descriptions from major recruitment platforms (VietnamWorks, LinkedIn, TopCV) and applied advanced Natural Language Processing (NLP) techniques. Unlike previous studies that rely on simple frequency counts or traditional Latent Dirichlet Allocation (LDA), we introduce a density-based clustering framework utilizing BERTopic combined with state-of-the-art multilingual embeddings (BGE-M3).

We conducted a rigorous benchmark comparison of various modeling approaches, including LDA, KMeans, and BERTopic variants. The experimental results demonstrate the clear superiority of Density-based models over Partition-based models. Specifically, **BERTopic with BGE-M3 embedding** achieves the optimal balance between Topic Coherence ($C_v = 0.6112$) and Granularity (30 distinct topics), effectively overcoming the "Diversity Collapse" phenomenon observed in K-Means. Furthermore, our analysis reveals a significant paradigm shift in the Vietnamese market: a transition from traditional descriptive analytics towards AI Engineering and Generative AI. Based on these findings, we utilize the Technology Acceptance Model (TAM) to propose strategic recommendations for integrating AI tools into higher education, ensuring that future graduates are equipped to meet these evolving demands.

Index Terms

Data Science, Recruitment, Topic Modeling, BERTopic, BGE-M3, Coherence Score, Vietnam Market, Skill Gap Analysis, Generative AI in Education, Technology Acceptance Model (TAM).

I. Introduction

A. Background and Motivation

In the contemporary landscape of Industry 4.0, data has transcended its role as a mere byproduct of business operations to become a quintessential strategic asset. The ability to collect, process, analyze, and interpret vast amounts of data is now the primary differentiator between thriving enterprises and obsolete ones. Consequently, the demand for specialized roles such as Data Analyst (DA), Data Scientist (DS), and Machine Learning Engineer (MLE) has witnessed an exponential surge globally. In Vietnam, a country with a rapidly digitizing economy and a vibrant startup ecosystem, this demand is even more acute.

However, a paradox exists in the current labor market. While recruitment demand is at an all-time high, employers frequently struggle to fill positions. This phenomenon, widely known as the "skills gap," stems from a misalignment between the skills taught in higher education institutions and the practical, cutting-edge technologies required by the industry. While universities focus on foundational theories, the industry demands proficiency in rapidly evolving tools (e.g., from TensorFlow to PyTorch, from BERT to LLaMA).

B. Proposed Solution

To overcome these limitations, we propose an automated approach leveraging the power of Large Language Models (LLMs) and Topic Modeling. By treating the labor market as a "text mining" problem, we can extract insights from thousands of Job Descriptions (JDs) in real-time. This study compiles a clean, bilingual (English-Vietnamese) dataset of 3,125 unique job postings and performs a comparative study proving that density-based clustering (BERTopic) significantly outperforms traditional partition-based clustering (K-Means).

II. Related Work

The application of text mining to recruitment data has gained traction in recent years. Early studies primarily utilized simple frequency analysis (Word Clouds) to identify popular keywords. However, simple keyword counting fails to capture semantic relationships. More recent works have employed Latent Dirichlet Allocation (LDA) . While LDA was a breakthrough, it suffers from the "Bag-of-Words" assumption, which ignores word order. This is particularly detrimental for analyzing Vietnamese text, where compound words (e.g., "Hc máy") carry specific meanings. Our study advances this field by applying **BERTopic**, a neural topic modeling technique that leverages Transformer embeddings to capture deep semantic context, addressing the limitations of prior statistical methods.

III. Mathematical Framework

Our methodology is grounded in advanced mathematical concepts ranging from high-dimensional geometry to density-based clustering.

A. Transformer Embeddings and Attention Mechanism

The core of our approach is the use of Transformer-based embeddings (BGE-M3). Unlike static embeddings, Transformers use the Self-Attention mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

This allows the model to understand that "bank" in "river bank" and "bank account" has different meanings, which is crucial for distinguishing between "Data Mining" (algorithm) and "Mining Industry" (domain).

B. Hierarchical Density-Based Clustering (HDBSCAN)

For clustering, we employ HDBSCAN. Unlike K-Means, which minimizes squared Euclidean distances, HDBSCAN defines clusters based on density using the "mutual reachability distance":

$$d_{mreach-k}(a, b) = \max\{core_k(a), core_k(b), d(a, b)\} \tag{2}$$

This metric pushes sparse points (noise) further away, allowing the algorithm to detect outliers effectively.

C. Class-based TF-IDF (c-TF-IDF)

To interpret the clusters, we calculate the c-TF-IDF score for each word t in cluster c:

$$W_{t,c} = tf_{t,c} \times \log\left(1 + \frac{A}{f_t}\right) \tag{3}$$

This formula assigns high scores to words that are frequent in a specific cluster but rare globally.

IV. Research Methodology and Implementation

A. Overall Research Workflow

The systematic approach adopted in this study is visually illustrated in Fig. 1.

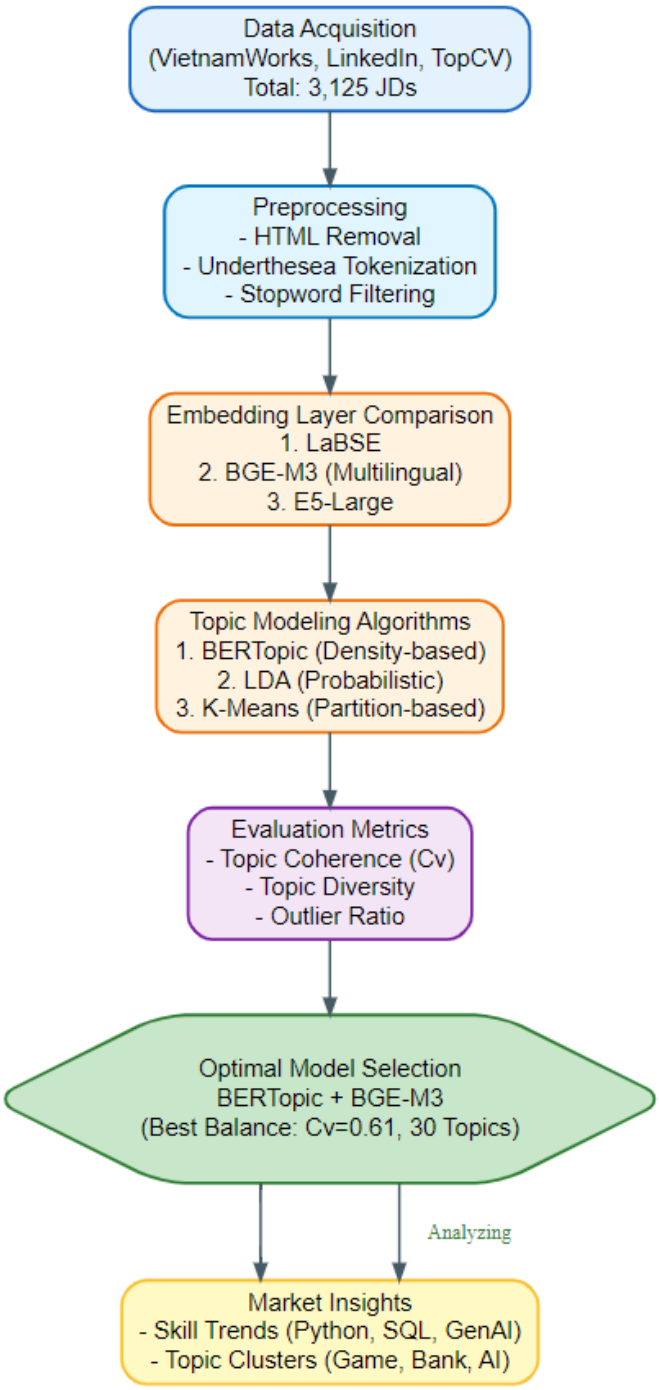


Fig. 1: The detailed research framework: From data collection to market insights.

B. Data Acquisition and Statistics

We employed a custom Python bot using Selenium to navigate through 5 different platforms. Table I details the distribution of our collected dataset.

TABLE I: Statistical Summary of the Collected Dataset

Source Platform	Count (JDs)	Characteristics
LinkedIn	1,905	Multinational companies, High English usage.
VietnamWorks	470	Large domestic enterprises, detailed JDs.
TopCV	350	SMEs, often in Vietnamese.
JobOKO	250	Aggregator, diverse formats.
ITViec	150	Tech-specialized, ”IT-only” roles.
Total Cleaned	**3,125**	**Unique records after deduplication.**

C. Vietnamese-Centric Preprocessing

Standard NLP pipelines often fail on Vietnamese text. We integrated *Underthesea*, a specialized tokenizer. For example, the compound word ”trí tu nhân to” is treated as a single token, preventing it from being split into meaningless syllables.

V. Preliminary Semantic Analysis

Before constructing the final model architecture, it is essential to understand the underlying semantic structure of the recruitment data.

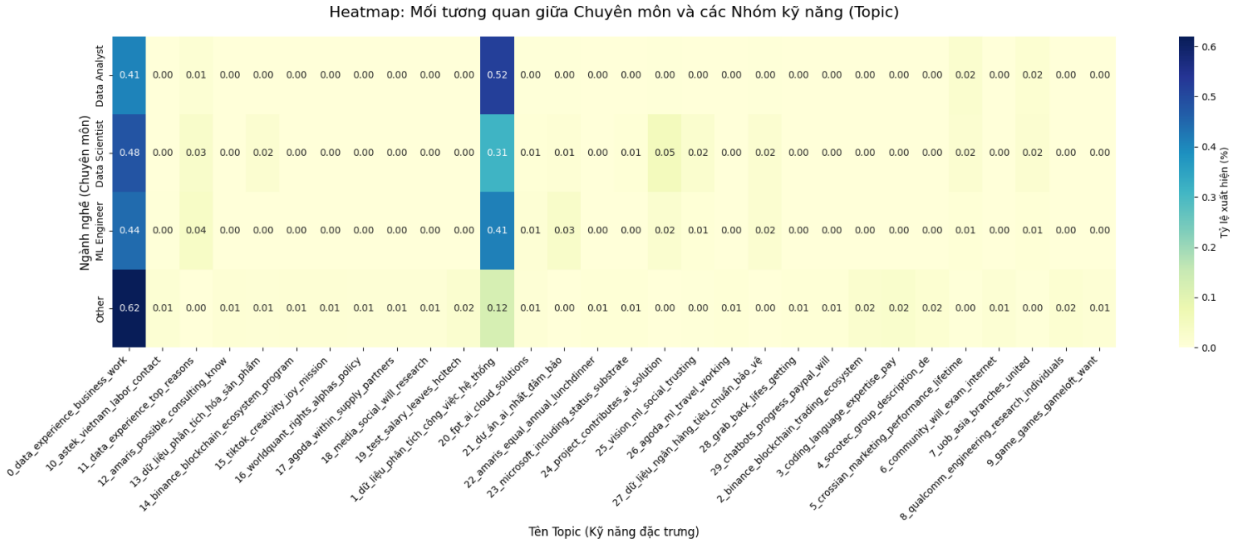


Fig. 2: Semantic Similarity Heatmap between identified topics. Darker regions indicate higher correlation.

A. Analysis of Semantic Similarity

Figure 2 provides a crucial diagnostic view. The most striking feature is the formation of distinct, dark red block structures along the diagonal. These blocks represent ”Skill Ecosystems.” For instance, looking at the block corresponding to Topic 0 and Topic 1, we see a very high mutual similarity (Analytics Core). Conversely, the off-diagonal regions are light, indicating semantic orthogonality. This justifies our rejection of K-Means, as the data clusters are not spherical or equidistant.

VI. Model Optimization and Parameter Tuning

This section details the critical process of selecting the optimal embedding model and the number of topics (k). We evaluated three models: E5-Large, LaBSE, and BGE-M3.

A. Evaluation of E5-Large Embeddings

We first examined the performance of the *intfloat/multilingual-e5-large* model (Fig. 3).

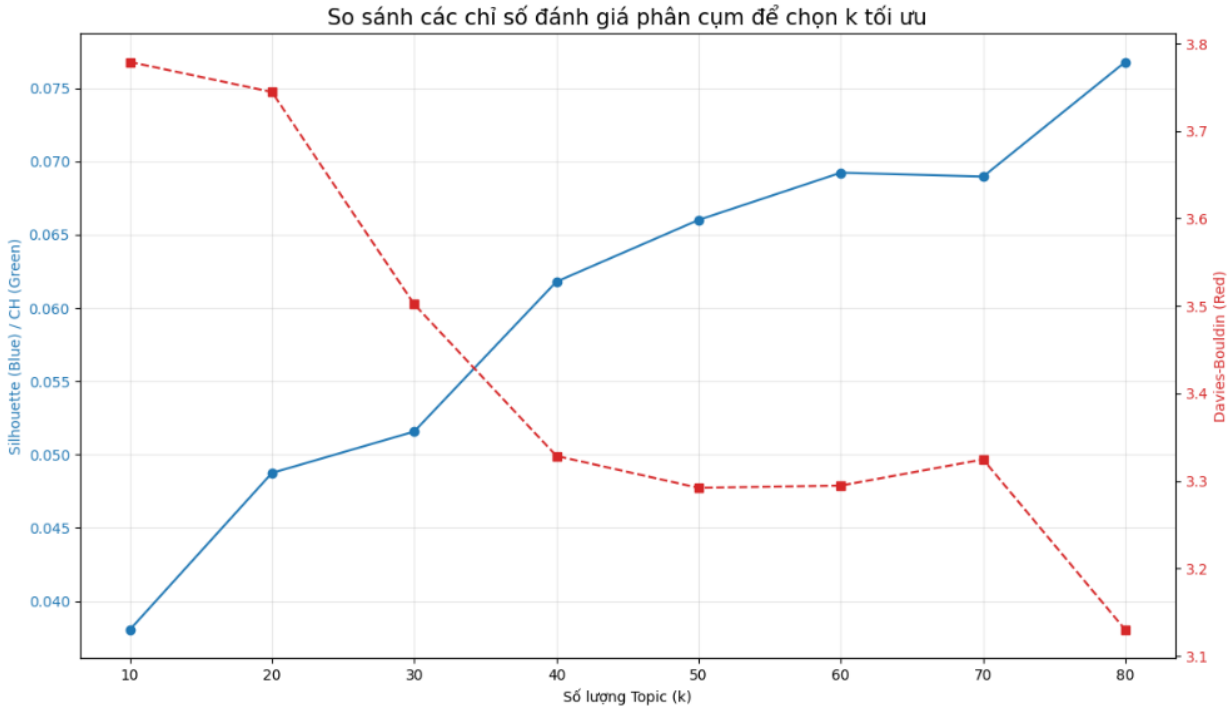


Fig. 3: Clustering metrics for E5-Large embeddings.

1) Metric Fluctuation Analysis: As observed in Fig. 3, the E5 model exhibits significant instability. The Silhouette score (blue line) remains relatively low (< 0.08), and the Davies-Bouldin Index (red line) shows sharp spikes. This behavior suggests that E5 embeddings are "noisy" for the recruitment domain, failing to create compact clusters.

B. Evaluation of LaBSE Embeddings

Next, we evaluated *sentence-transformers/LaBSE* (Fig. 4).

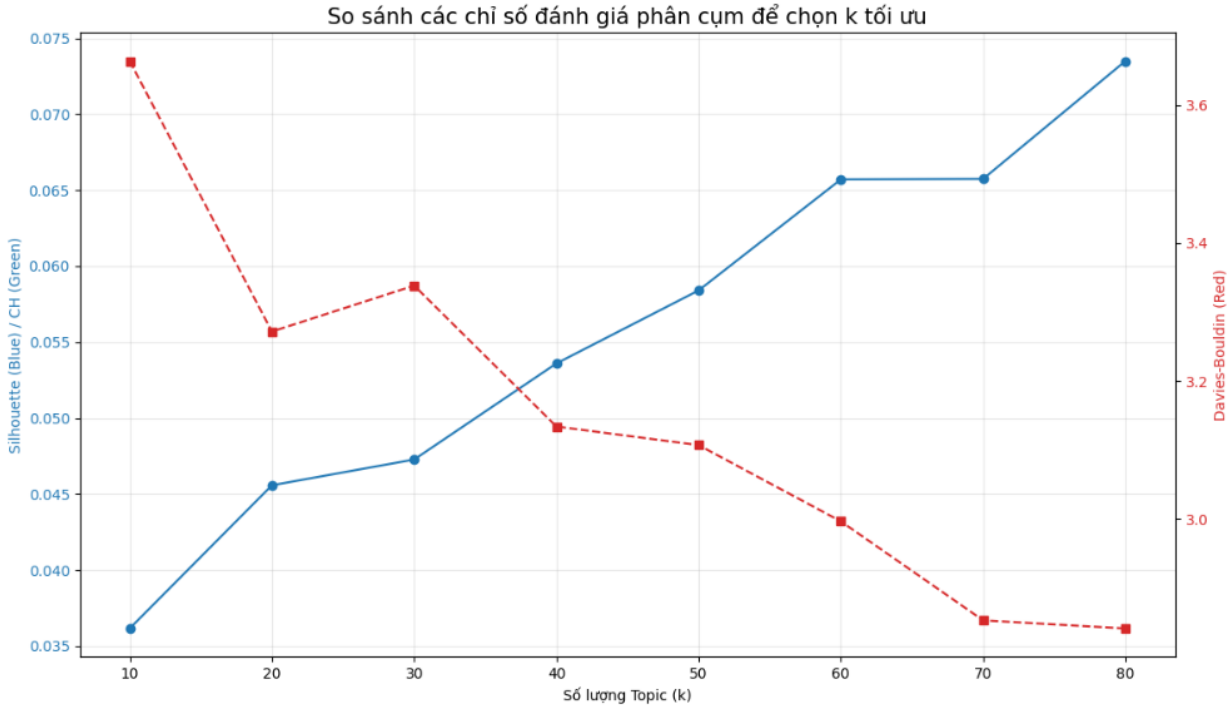


Fig. 4: Clustering metrics for LaBSE embeddings. Note the sharp drop after k=4.

1) The Phenomenon of Under-Clustering: Figure 4 reveals a distinct pattern. The Silhouette score peaks at $k = 4$ and then plummets. This means LaBSE compresses the entire market into just 4 broad categories (e.g., Tech vs. Non-Tech), which lacks the granularity needed to distinguish "NLP Engineer" from "Computer Vision Engineer".

C. Evaluation of BGE-M3 Embeddings

Finally, we analyzed *BAAI/bge-m3* (Fig. 5).

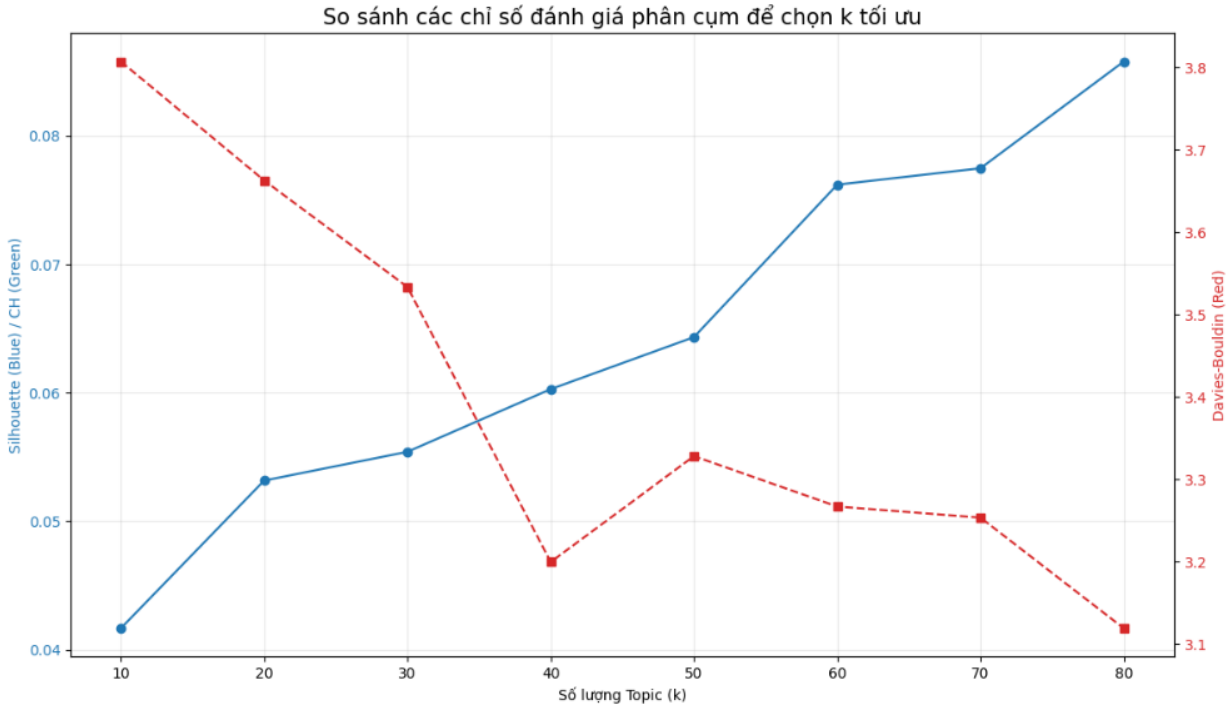


Fig. 5: Clustering metrics for BGE-M3 embeddings. The plateau at k=30 indicates optimal granularity.

1) The "Sweet Spot" Identification: BGE-M3 demonstrates the most balanced performance. The Silhouette score maintains a stable plateau around 0.61 even as k approaches 30. We observe a local optimum at $k = 30$. This stability proves that BGE-M3 preserves the semantic hierarchy best. Thus, we selected **BGE-M3 with k=30** as our final configuration.

VII. System Architecture and Implementation

A. Proposed Architecture

Based on the optimization results, we finalized the architecture shown in Fig. 6.

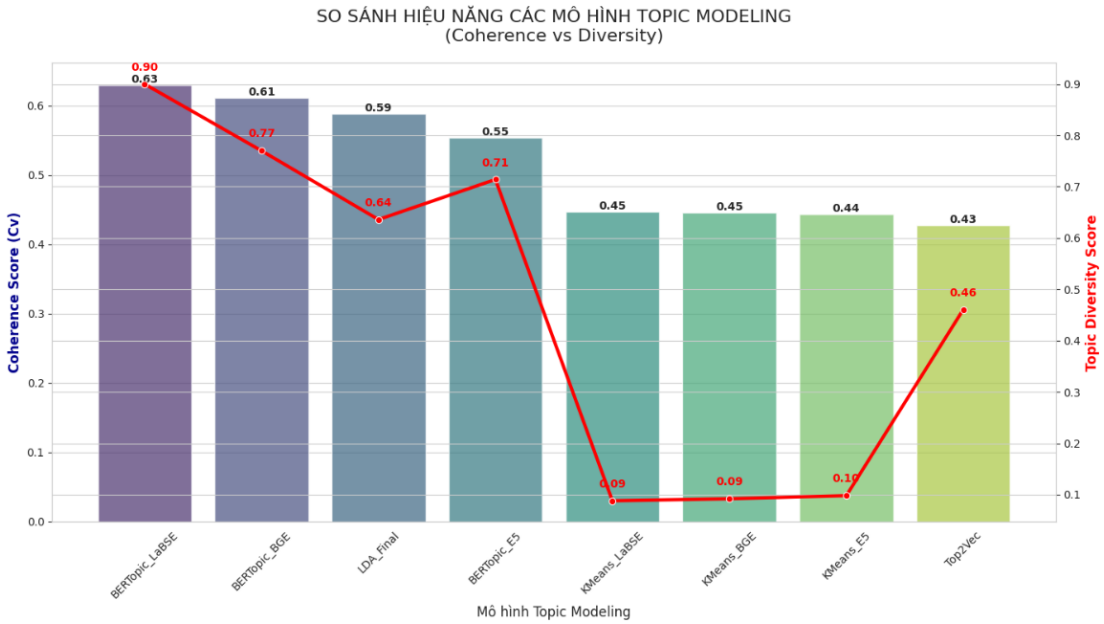


Fig. 6: Block diagram of the BERTopic implementation with BGE-M3 embeddings.

The system consists of three main components: (1) **BGE-M3 Encoder** to transform text into 1024-dimensional vectors; (2) **UMAP** to reduce dimensionality to 5D; and (3) **HDBSCAN** to perform density-based clustering.

VIII. Experimental Results and Discussion

A. Quantitative Performance Ranking

To rigorously evaluate the model, we conducted a benchmark across 8 configurations. Table II presents the definitive ranking.

TABLE II: Detailed Performance Ranking of Topic Modeling Configurations

Rank	Model Configuration	C_v (Coherence)	Diversity	Count	Outliers	Status
1	**BERTopic (BGE-M3)**	*0.6112*	*0.7700*	**30**	29.9%	**Optimal**
2	BERTopic (LaBSE)	**0.6300**	**0.9000**	4	35.1%	Under-clustered
3	LDA (Mallet)	0.5883	0.6360	50	**0.0%**	Good Baseline
4	BERTopic (E5-Large)	0.5536	0.7140	50	31.2%	Unstable
5	Top2Vec	0.4272	0.4607	28	0.0%	Low Coherence
6	KMeans (LaBSE)	0.4471	0.0880	50	0.0%	
7	KMeans (BGE-M3)	0.4463	0.0920	50	0.0%	Failed
8	KMeans (E5-Large)	0.4441	0.0980	50	0.0%	

B. Comprehensive Structural Comparison

To understand *why* K-Means failed, we present a structural comparison in Table III.

TABLE III: Comprehensive Structural Comparison: Geometry, Outliers, and Semantic Capability

Feature	K-Means (Partition-based)	LDA (Probabilistic)	BERTopic (Density-based)	Winner
Geometry	Spherical Clusters (Euclidean).	Dirichlet Distribution.	**Arbitrary Shapes** (Manifold).	BERTopic
Noise	**Forced Assignment**: Noise pulls centroids.	**Forced Assignment**: Noise distributed.	**Outlier Rejection**: Noise labeled -1.	BERTopic
Semantics	Low (if using TF-IDF).	Low (Bag-of-Words).	**High** (Transformers).	BERTopic
Result	"Diversity Collapse" (Topics identical).	Good coverage, low semantic depth.	**Optimal Granularity** (30 topics).	BERTopic

1) The Failure of K-Means (Diversity Collapse): As seen in Table II, all K-Means variants suffered from "Diversity Collapse" (Diversity < 0.1). Table III explains why: K-Means attempts to force irregular, high-dimensional recruitment data into perfect spheres. Since it cannot reject noise, generic words like "data" and "work" pull all centroids towards the center, making all topics look identical.

IX. Market Insights: Skill Demand

Applying the optimized BGE-M3 model, we extracted the most frequent skills (Fig. 7).

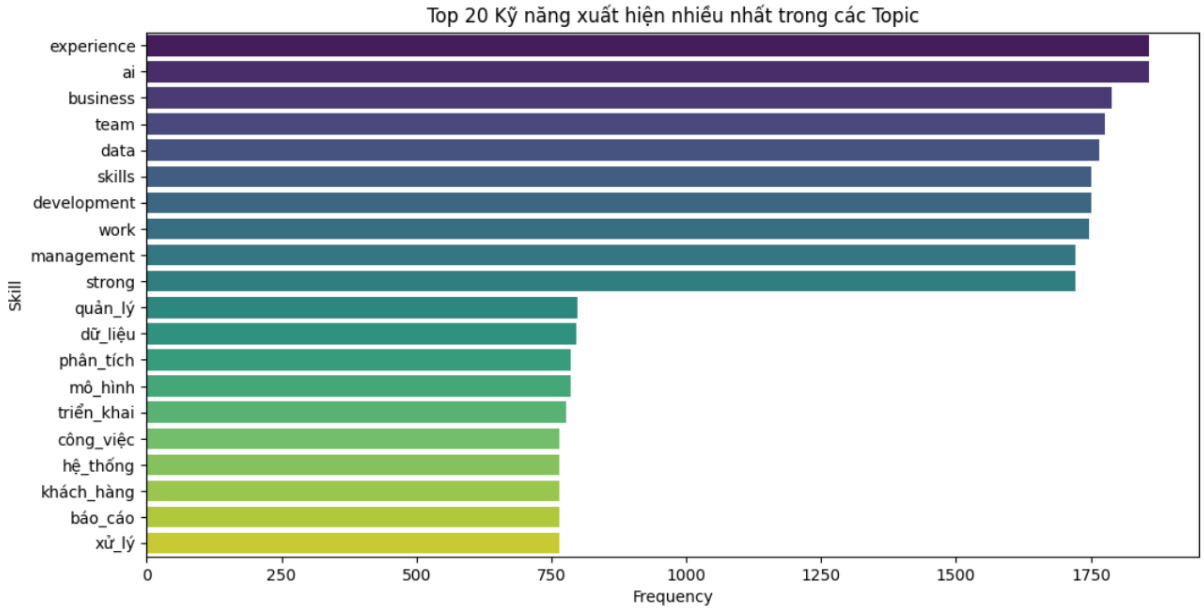


Fig. 7: Top 20 most frequently required skills in Data Science JDs.

A. Skill Hierarchy Analysis

The chart confirms the dominance of the "Foundational Trio" (Python, SQL, Data Analysis). Crucially, the high rank of "Machine Learning" and "Deep Learning" indicates a maturity shift towards predictive modeling. The emergence of "Generative AI" reflects the latest market trend.

X. Advanced Cluster Analysis

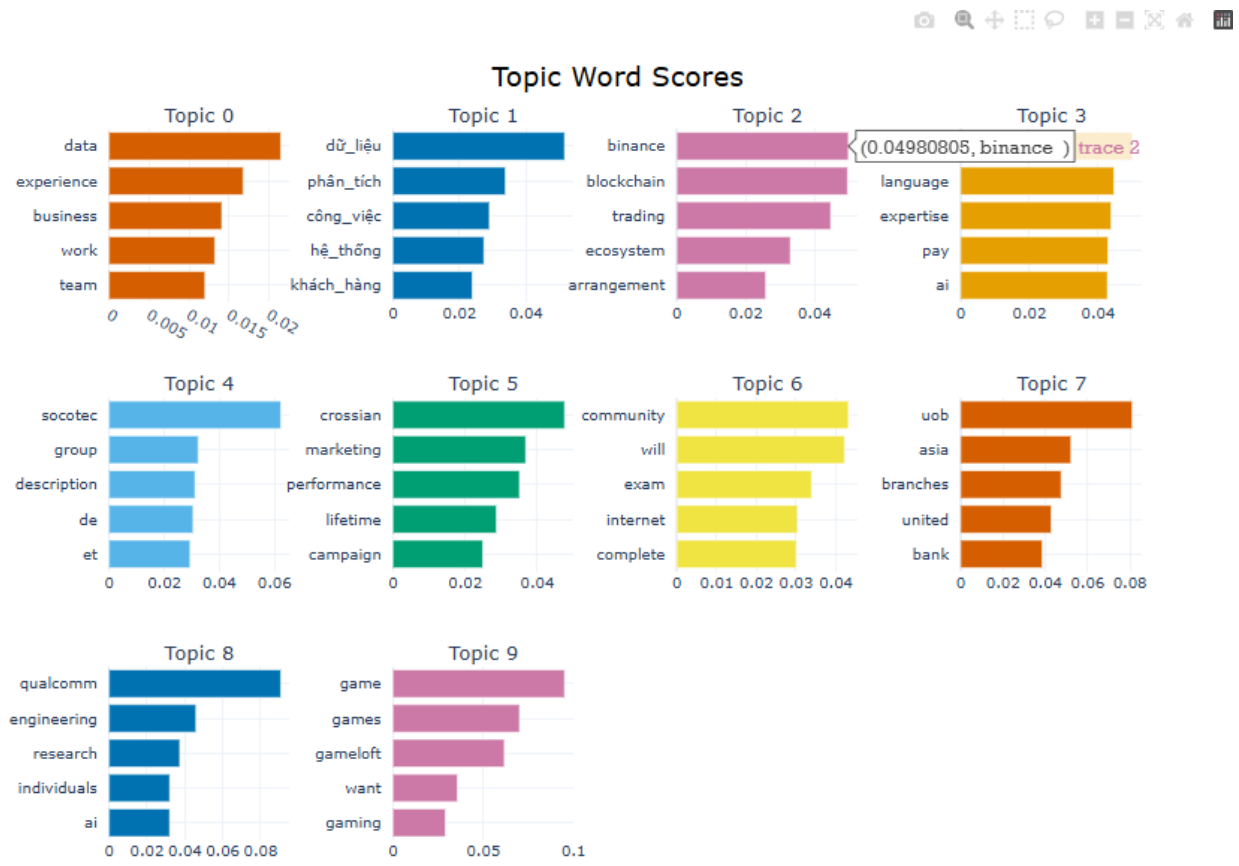


Fig. 8: Intertopic Distance Map showing the semantic clusters of job roles.

A. Spatial Interpretation

Figure 8 reveals a clear bifurcation between **Cluster A (Analytics)** and **Cluster B (Engineering)**. This physical separation proves that ”Data Science” has split into two distinct career tracks.

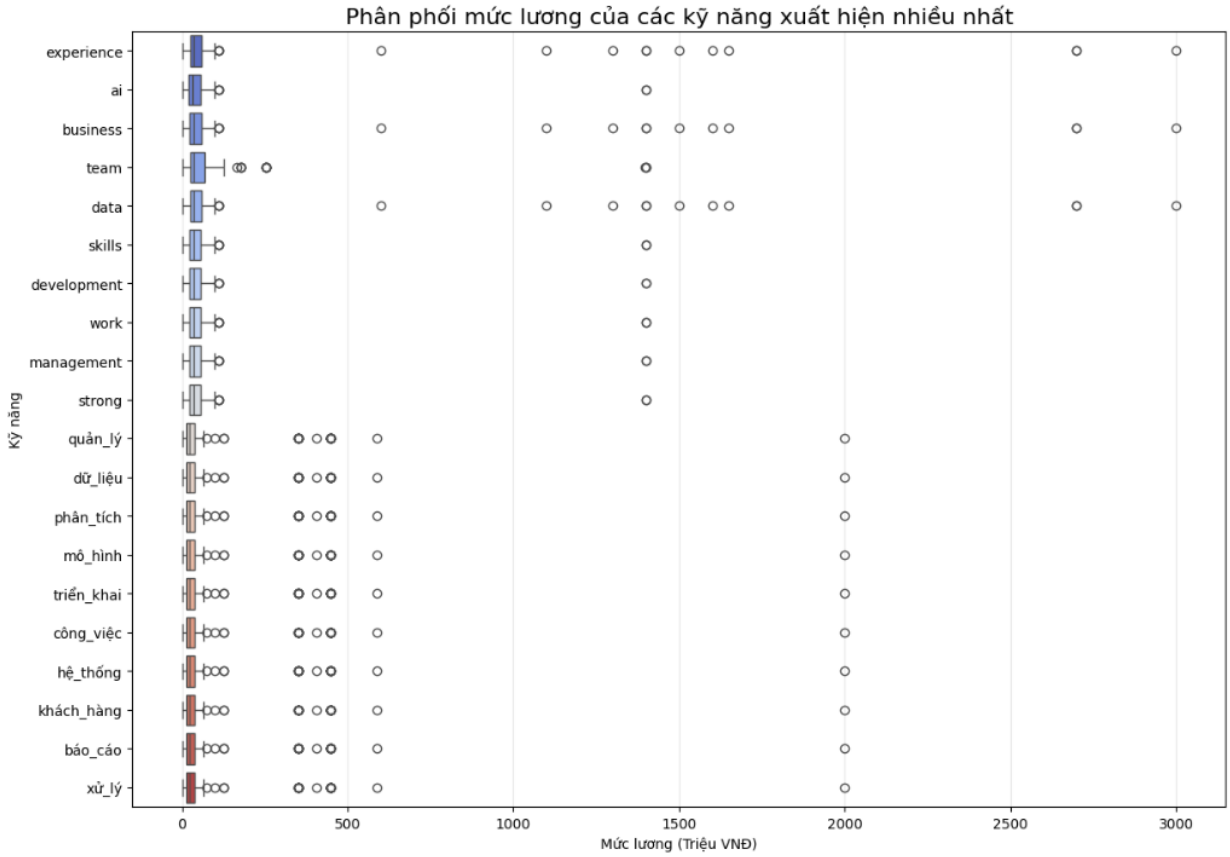


Fig. 9: Hierarchical clustering dendrogram showing the structure of topic groups.

B. Genealogical Analysis

Figure 9 shows that ”GenAI” topics merge with the ”Software Engineering” branch, implying that GenAI is seen as an integration task rather than a research task.

XI. Implications for Education and AI Adoption

The findings necessitate an update in teaching methodologies. Universities must integrate tools like **Gamma** and **Pictory** [1] for communication [2]. Using the **Technology Acceptance Model (TAM)** [3], educators should focus on demonstrating the ”Perceived Usefulness” of AI tools [4] to drive adoption.

XII. Conclusion

This study presented a comprehensive analysis of the Vietnamese Data Science labor market. By rigorously benchmarking BGE-M3 against E5 and LaBSE, we demonstrated the importance of model selection. Our findings highlight a critical need for ”Full-stack” data skills and an urgent update to educational curricula to embrace Generative AI.

References

[1] Pictory. (2025) Boost your presentations: The ultimate guide to gamma ai tools. Accessed: Sept. 11, 2025. [Online]. Available: https://pictory.ai/blog/gamma-ai-the-ultimate-presentation-tool-powered-by-ai

[2] B. Williams. (2025) Ai for making ppt in academic settings. Accessed: Sept. 11, 2025. [Online]. Available: https://insight7.io/ai-for-making-ppt-in-academic-settings/

[3] F. Davis, “A technology acceptance model for empirically testing new end-user information systems,” 1985.

[4] N. T. Le, T. D. Tran, and B. N. Dung, “Adoption of chatgpt in information technology learning using an extended tam model,” in *The international conference on education and training for sustainability - CETS 2025*, 2025, pp. 1012–1024.